

Erdős Problem #12

EP-12

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let A be an infinite set such that there are no distinct $a, b, c \in A$ such that $a \mid (b + c)$ and $b, c > a$. Is there such an A with $\liminf \frac{|A \cap \{1, \dots, N\}|}{N^{1/2}} > 0$? Does there exist some absolute constant $c > 0$ such that there are always infinitely many N with $|A \cap \{1, \dots, N\}| < N^{1-c}$? Is it true that $\sum_{n \in A} \frac{1}{n} < \infty$?

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